

Assignment 4 — Write-up

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Part A — Sparse Linear Predictors and Model Selection

Throughout, $\mathcal{X} = \mathbb{R}^d$, $\mathcal{Y} = \{-1, +1\}$, $\mathcal{H}_k = \{x \mapsto \text{sign}(\langle w, x \rangle) : \|w\|_0 \leq k\}$, $\mathcal{H} = \bigcup_{k=1}^d \mathcal{H}_k$, and $S \sim \mathcal{D}^n$.

A.1 VC dimension through supports

For a fixed support $I \subseteq [d]$ with $|I| = k$, write

$$\mathcal{H}_I = \{x \mapsto \text{sign}(\langle w, x \rangle) : \text{supp}(w) \subseteq I\}.$$

The map $w \mapsto w|_I$ identifies $\mathcal{H}_I$ with the class of homogeneous halfspaces in $\mathbb{R}^k$, applied to the coordinates of x indexed by I . Homogeneous halfspaces in $\mathbb{R}^k$ have VC dimension k , so by Sauer–Shelah

$$\Gamma_{\mathcal{H}_I}(n) \leq \left(\frac{en}{k}\right)^k \quad (n \geq k).$$

Since $\mathcal{H}_k = \bigcup_{|I|=k} \mathcal{H}_I$, a union bound on dichotomies gives

$$\Gamma_{\mathcal{H}_k}(n) \leq \binom{d}{k} \left(\frac{en}{k}\right)^k \leq \left(\frac{ed}{k}\right)^k \left(\frac{en}{k}\right)^k = \left(\frac{e^2 dn}{k^2}\right)^k.$$

If $\text{VCdim}(\mathcal{H}_k) = m$, then $2^m \leq \Gamma_{\mathcal{H}_k}(m) \leq (e^2 dm/k^2)^k$, so $m \leq k \log_2(e^2 dm/k^2)$. A standard lemma (e.g. Shalev-Shwartz–Ben-David, Lemma A.2) converts $m \leq k \log_2(am/k)$ into $m \leq 2k \log_2(2ak/k) = O(k \log(ed/k))$. Hence

$\text{VCdim}(\mathcal{H}_k) = O(k \log(ed/k)).$

Lower-bound sanity check. $\mathcal{H}_k$ contains $\mathcal{H}_I$ for any single I , and $\text{VCdim}(\mathcal{H}_I) = k$; so $\text{VCdim}(\mathcal{H}_k) \geq k$. The upper bound is tight up to the $\log(ed/k)$ factor, which is the price of choosing the support.

A.2 Penalty-based SRM

Class-level SRM theorem (from lecture). Let $\mathcal{H} = \bigcup_{k=1}^d \mathcal{H}_k$ with $p_k > 0$, $\sum_k p_k \leq 1$. Suppose for each k there is a uniform-convergence rate $\varepsilon_k(n, \delta)$ such that, with probability $\geq 1 - \delta$ over S , $\sup_{h \in \mathcal{H}_k} |L_S(h) - L_{\mathcal{D}}(h)| \leq \varepsilon_k(n, \delta)$. Define $\text{pen}(k, n, \delta) = \varepsilon_k(n, p_k \delta)$ and $\hat{h} \in \arg \min_{k, h \in \mathcal{H}_k} \{L_S(h) + \text{pen}(k, n, \delta)\}$. Then with probability $\geq 1 - \delta$,

$$L_{\mathcal{D}}(\hat{h}) \leq \inf_{k, h \in \mathcal{H}_k} \{L_{\mathcal{D}}(h) + 2 \text{pen}(k, n, \delta)\}.$$

Instantiation for $\mathcal{H}_k$. By A.1, $\text{VCdim}(\mathcal{H}_k) = d_k = O(k \log(ed/k))$. The VC uniform-convergence theorem (e.g. Vapnik–Chervonenkis / Massart) gives

$$\varepsilon_k(n, \delta) \leq C_0 \sqrt{\frac{d_k + \log(1/\delta)}{n}} \leq C_1 \sqrt{\frac{k \log(ed/k) + \log(1/\delta)}{n}}.$$

Apply with confidence parameter $p_k\delta$:

$$\text{pen}(k, n, \delta) \leq C \sqrt{\frac{k \log(ed/k) + \log(1/p_k) + \log(1/\delta)}{n}}.$$

By the SRM theorem, with probability $\geq 1 - \delta$,

$$L_{\mathcal{D}}(\hat{h}) \leq \inf_{1 \leq k \leq d, h \in \mathcal{H}_k} \left\{ L_{\mathcal{D}}(h) + C \sqrt{\frac{k \log(ed/k) + \log(1/p_k) + \log(1/\delta)}{n}} \right\}.$$

Interpretation of each term.

- $k \log(ed/k)$ is the **support cost**: it is the VC dimension of $\mathcal{H}_k$, and it pays for uniform convergence *within* the level k — i.e. for the freedom to choose which k coordinates carry weight and which halfspace to put on them.
- $\log(1/p_k)$ is the **sparsity-level cost**: it is the price of the union bound across the d sparsity levels. With $p_k \propto 1/k^2$ we have $\log(1/p_k) = 2 \log k + O(1)$, much smaller than $\log(d)$ for the levels we actually care about.
- $\log(1/\delta)$ is the usual confidence cost.

A.3 Comparison with dense halfspaces

Suppose w^* with $\|w^*\|_0 \leq s$ and $L_{\mathcal{D}}(w^*) \leq \eta$. Plugging $h = \text{this } w^*$ and $k = s$ into the SRM bound and forcing the penalty $\leq \varepsilon$:

$$n \geq C \frac{s \log(ed/s) + \log(1/p_s) + \log(1/\delta)}{\varepsilon^2} = O\left(\frac{s \log(ed/s) + \log(1/\delta)}{\varepsilon^2}\right)$$

(using $p_s \propto 1/s^2$, so $\log(1/p_s) = O(\log s)$, absorbed into $\log(ed/s)$).

Dense halfspaces in $\mathbb{R}^d$ have VC dimension d , so the corresponding sample complexity is

$$n_{\text{dense}} = O\left(\frac{d + \log(1/\delta)}{\varepsilon^2}\right).$$

When sparse is smaller. Sparse beats dense when $s \log(ed/s) \ll d$, i.e. roughly when $s = o(d/\log d)$. For $s = d$ the bounds are identical up to a log factor; for $s = \Theta(\log d)$ the sparse bound is exponentially better in d .

A.4 Validation as model selection

Split S evenly into S_1 (training, size $m = n/2$) and S_2 (validation, size m). Define

$$w_k \in \arg \min_{\|w\|_0 \leq k} L_{S_1}(w), \quad \hat{k} \in \arg \min_{1 \leq k \leq d} L_{S_2}(w_k).$$

Step 1 (validation step). Condition on S_1 . The list $w_1, \dots, w_d$ is then a *fixed* (i.e. S_2 -independent) set of d classifiers. Apply Hoeffding plus a union bound over these d candidates on the independent sample S_2 of size $m = n/2$: with probability $\geq 1 - \delta/2$, for all $k \in \{1, \dots, d\}$,

$$|L_{S_2}(w_k) - L_{\mathcal{D}}(w_k)| \leq \sqrt{\frac{\log(4d/\delta)}{2m}} \leq C_2 \sqrt{\frac{\log(d/\delta)}{n}}.$$

Therefore, on this event,

$$L_{\mathcal{D}}(w_{\hat{k}}) \leq L_{S_2}(w_{\hat{k}}) + C_2 \sqrt{\frac{\log(d/\delta)}{n}} \leq L_{S_2}(w_k) + C_2 \sqrt{\frac{\log(d/\delta)}{n}} \quad \forall k,$$

and combining the two Hoeffding inequalities,

$$L_{\mathcal{D}}(w_{\hat{k}}) \leq \min_{1 \leq k \leq d} L_{\mathcal{D}}(w_k) + 2C_2 \sqrt{\frac{\log(d/\delta)}{n}}. \quad (\star)$$

Step 2 (training step). For each fixed k , w_k minimizes empirical risk over $\mathcal{H}_k$ on S_1 . By the VC uniform-convergence bound applied to $\mathcal{H}_k$ with confidence $\delta/(2d)$ and sample size $m = n/2$, plus a union bound over k , with probability $\geq 1 - \delta/2$, for every k and every w with $\|w\|_0 \leq k$,

$$L_{\mathcal{D}}(w_k) \leq L_{\mathcal{D}}(w) + C_3 \sqrt{\frac{k \log(ed/k) + \log(d/\delta)}{n}}. \quad (\star\star)$$

Combine. Union bound the two events. With probability $\geq 1 - \delta$, taking the infimum over k and w in $(\star)$ via $(\star\star)$,

$$L_{\mathcal{D}}(w_{\hat{k}}) \leq \inf_{1 \leq k \leq d, \|w\|_0 \leq k} \left\{ L_{\mathcal{D}}(w) + C \sqrt{\frac{k \log(ed/k) + \log(d/\delta)}{n}} \right\}.$$

Comparison with penalty SRM.

	Penalty-SRM	Validation
Sample used to <i>fit</i> candidates	all n	$n/2$
Sample used to <i>choose</i> k	the same n	a separate, independent $n/2$
Complexity term for support	$k \log(ed/k)$	$k \log(ed/k)$
Complexity term for choosing k	$\log(1/p_k) = O(\log k)$	$\log d$

Validation pays a constant factor of ~ 2 in n (the split) and replaces the prior-dependent $\log(1/p_k)$ by $\log d$. In exchange, the validation step is purely finite-class and never invokes the (in general computationally hard) penalty.

A.5 Computation

Brute-force algorithm (realizable case). Enumerate every support $I \subseteq [d]$ with $|I| = k$. For each I , solve the homogeneous-halfspace feasibility problem

$$\text{find } w \in \mathbb{R}^I : \quad y_i \langle w, x_i|_I \rangle \geq 0 \quad (i = 1, \dots, n)$$

(strict-inequality / margin form, e.g. by LP or Perceptron). This is solvable in $\text{poly}(k, n)$ time. If any support yields a feasible w , output it; the realizable assumption guarantees at least one such I exists.

Runtime. $\binom{d}{k} \cdot \text{poly}(k, n) \leq (ed/k)^k \cdot \text{poly}(k, n)$. This is polynomial in d iff $k = O(1)$ (a constant). For $k = \Theta(\log d)$ already, it is $d^{\Theta(\log d)}$, superpolynomial.

Statistical vs computational complexity. Part A.3 shows that $n = \tilde{O}(s/\varepsilon^2)$ samples suffice to learn an s -sparse halfspace, with *no* explicit dependence on d beyond a log. But the brute-force algorithm needs d^s time. So small sample complexity does not buy small runtime: choosing the support is a combinatorial problem whose decision version (PAC-learning sparse halfspaces under adversarial noise) is NP-hard in general. This is exactly the Week 4 picture — VC theory says “the information is in the sample”; computation says “extracting it can still be hard.”

Part B — PAC-Bayes for Thresholds

$\mathcal{X}_N = \{1, \dots, N\}$, $\mathcal{Y} = \{0, 1\}$, $\mathcal{H}_N = \{h_t : t \in \{1, \dots, N+1\}\}$ with $h_t(x) = \mathbf{1}[x \geq t]$.

B.1 VC dimension and point-posterior PAC-Bayes

VCdim = 1. A single point x_0 is shattered: $h_{x_0}(x_0) = 1$ and $h_{x_0+1}(x_0) = 0$ (taking h_{N+1} if $x_0 = N$). For *any* two points $x_1 < x_2$, the labeling $(y_1, y_2) = (1, 0)$ is unachievable: $h_t(x_1) = 1 \Rightarrow t \leq x_1 < x_2 \Rightarrow h_t(x_2) = 1 \neq 0$. Hence $\text{VCdim}(\mathcal{H}_N) = 1$.

KL for a point posterior. P uniform on $N + 1$ thresholds, $Q_S = \delta_{h_i}$:

$$\text{KL}(Q_S \| P) = \log \frac{1}{P(h_i)} = \log(N + 1).$$

PAC-Bayes bound. In the realizable setting, $L_S(Q_S) = L_S(h_i) = 0$, so

$$L_{\mathcal{D}}(\delta_{h_i}) \leq \sqrt{\frac{\log(N + 1) + \log(2n/\delta)}{2(n - 1)}}.$$

Why the $\log(N + 1)$. A VC realizable bound for thresholds gives the much sharper $O(\log(1/\delta)/n)$ rate with *no* $\log N$. The $\log(N + 1)$ in PAC-Bayes is the KL of a point mass against a uniform prior over $N + 1$ atoms: the prior is fixed before seeing the data and is forced to spread its mass across all candidate thresholds, so concentrating onto the empirical winner costs $\log(N + 1)$ nats of KL. This is a property of the *certificate* (specifically: of using a point posterior with a prior that is uniform over an N -dependent set), not of the underlying learning problem.

B.2 Version-space posterior

Version space. If $y_i = 0$, consistency requires $h_t(x_i) = 0$, i.e. $x_i < t$, i.e. $t > x_i$. Hence $t > a(S) := \max_{i: y_i=0} x_i$ (with $a(S) = 0$ if no negatives). If $y_i = 1$, consistency requires $x_i \geq t$, i.e. $t \leq x_i$. Hence $t \leq b(S) := \min_{i: y_i=1} x_i$ (with $b(S) = N+1$ if no positives). So

$$V(S) = \{t : a(S) < t \leq b(S)\}, \quad |V(S)| = b(S) - a(S).$$

This set is nonempty in the realizable case since the true threshold lies in it.

KL of Q_V . Q_V uniform on $V(S)$, P uniform on $N + 1$ thresholds, so

$$\text{KL}(Q_V \| P) = \sum_{t \in V} \frac{1}{|V|} \log \frac{1/|V|}{1/(N+1)} = \log \frac{N+1}{|V(S)|}.$$

Empirical risk. For every $t \in V(S)$, h_t is consistent with S , so $L_S(h_t) = 0$; therefore $L_S(Q_V) = \mathbb{E}_{t \sim Q_V} L_S(h_t) = 0$.

Why Q_V can beat the point posterior. Both achieve $L_S = 0$, but $\text{KL}(Q_V \| P) = \log(N+1) - \log|V(S)|$, smaller than $\log(N+1)$ by exactly $\log|V(S)|$. When the version space is large (typical for small n), this gives a genuinely tighter PAC-Bayes certificate.

B.3 Spreading helps KL, but can hurt true risk

True risk of Q_W . The marginal of $\mathcal{D}$ on $\mathcal{X}_N$ is uniform and labels come from h_τ . For any t ,

$$L_{\mathcal{D}}(h_t) = \Pr_x[h_t(x) \neq h_\tau(x)] = \frac{|\{x \in [N] : (x \geq t) \oplus (x \geq \tau)\}|}{N} = \frac{|t - \tau|}{N},$$

since the disagreement set is the integer interval between $\min(t, \tau)$ and $\max(t, \tau) - 1$, of size $|t - \tau|$. Hence

$$L_{\mathcal{D}}(Q_W) = \mathbb{E}_{t \sim Q_W} L_{\mathcal{D}}(h_t) = \frac{1}{N|W|} \sum_{t \in W} |t - \tau|.$$

Concrete example. Take $N = 20$, $\tau = 11$, and a one-point sample $S = ((1, 0))$. Then $a(S) = 1$, $b(S) = N+1 = 21$, so $V(S) = \{2, 3, \dots, 21\}$ with $|V(S)| = 20$ — large.

- $\text{KL}(\delta_{h_\tau} \| P) = \log 21 \approx 3.04$.
- $\text{KL}(Q_{V(S)} \| P) = \log \frac{21}{20} \approx 0.0488$. So $\text{KL}(Q_V \| P) < \text{KL}(\delta_{h_\tau} \| P)$.
- $L_{\mathcal{D}}(\delta_{h_\tau}) = 0$ (h_τ is the true labeller).
- $L_{\mathcal{D}}(Q_V) = \frac{1}{20 \cdot 20} \sum_{t=2}^{21} |t - 11| = \frac{1}{400} (\sum_{j=1}^9 j + 0 + \sum_{j=1}^{10} j) = \frac{45+0+55}{400} = \frac{100}{400} = 0.25$. So $L_{\mathcal{D}}(Q_V) > L_{\mathcal{D}}(\delta_{h_\tau})$.

No contradiction with PAC-Bayes. PAC-Bayes gives an *upper* bound on $L_{\mathcal{D}}(Q)$, not a monotone equivalence: a smaller KL produces a smaller *bound*, but smaller bound does not imply smaller actual risk. Both certificates correctly upper-bound their respective $L_{\mathcal{D}}$ values; spreading the posterior across the version space gives a tighter *certificate* even when it gives a *worse* posterior, because the posterior is chosen to make $\text{KL} + L_S$ small, not $L_{\mathcal{D}}$ small (which the learner does not see).

B.4 Every fixed prior can be attacked

Pigeonhole. P is a probability distribution, so $\sum_{\tau=2}^N P(h_\tau) \leq 1$. Since this is a sum of $N-1$ nonnegative terms, at least one satisfies

$$P(h_\tau) \leq \frac{1}{N-1} \quad \text{for some } \tau \in \{2, \dots, N\}. \quad (\dagger)$$

Fix this τ .

$\mathcal{D}_\tau$ is realizable. Under $\mathcal{D}_\tau$, $\Pr[(x, y) = (\tau-1, 0)] = \Pr[(x, y) = (\tau, 1)] = 1/2$. Both points are correctly labelled by h_τ : $h_\tau(\tau-1) = 0$, $h_\tau(\tau) = 1$. So $L_{\mathcal{D}}(h_\tau) = 0$.

Both support points appear. For $S \sim \mathcal{D}_\tau^n$, let A = event that some $x_i = \tau-1$ and some $x_j = \tau$. Then

$$\Pr[A^c] \leq \Pr[\text{all } x_i = \tau-1] + \Pr[\text{all } x_i = \tau] = 2 \cdot (1/2)^n = 2^{1-n},$$

so $\Pr[A] \geq 1 - 2^{1-n}$.

Version space collapses to $\{\tau\}$. On A , $a(S) = \tau - 1$ (the only label-0 value in the support) and $b(S) = \tau$ (the only label-1 value), so by B.2, $V(S) = \{t : \tau - 1 < t \leq \tau\} = \{\tau\}$.

Forced posterior. Any posterior Q with $L_S(Q) = 0$ is supported on $V(S) = \{\tau\}$, hence $Q = \delta_{h_\tau}$. Then

$$\text{KL}(Q\|P) = \log \frac{1}{P(h_\tau)} \stackrel{(\dagger)}{\geq} \log(N-1).$$

What this does/does not show.

- *Does show:* For any fixed prior P over $\mathcal{H}_N$, there exists a realizable distribution on which every zero-empirical-error PAC-Bayes certificate must incur $\text{KL} \geq \log(N-1)$. So the *bound* it produces is at least $\sqrt{(\log(N-1) + \log(2n/\delta))/(2(n-1))}$. This is a $\log N$ lower bound on the PAC-Bayes *certificate*.
- *Does not show:* A lower bound on the sample complexity of *learning* thresholds. ERM on thresholds achieves $L_{\mathcal{D}}(\hat{h}) = O(\log(1/\delta)/n)$ uniformly in N (VC dimension 1). The certificate is loose; the learner is fine. The slack lives in the choice to (a) use a *fixed* prior chosen before seeing the data and (b) insist on a *zero-empirical-error* posterior. Either relaxing the $L_S(Q) = 0$ requirement (KL-bound trades off against empirical risk) or allowing a data-dependent prior (e.g. localized PAC-Bayes) can sidestep the bound.